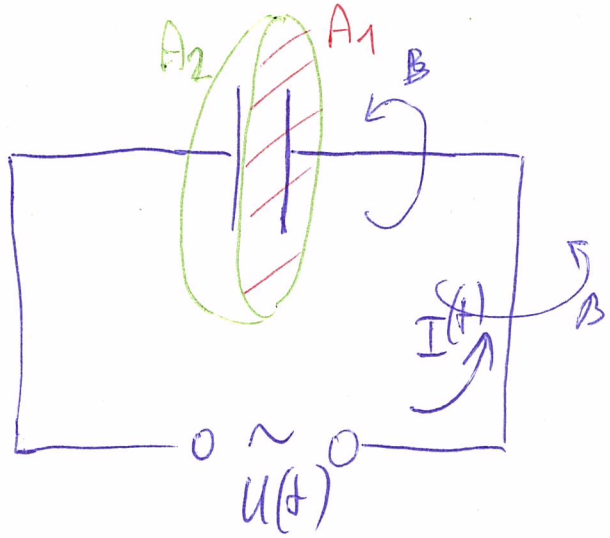
13. Displacement current, Maxwell's eq.s

Let us have a closer look on Ampère's law!
For a time-variant case.



We investigate the magnetic field inside a capacitor being charged.

$$\oint \underline{B}\,d\underline{r} = \mu_0 \sum I = \mu_0 \iint_A \underline{j}\,d\underline{c}$$

$$\mu_0 \iint_{A_1} \underline{j}\,d\underline{s} = 0 \Rightarrow B = 0$$

$$\mu_0 \iint_{A_2} \underline{j}\,d\underline{s} = I \Rightarrow B \neq 0$$

$\Downarrow\Downarrow$

$\ast\ast$ Contradiction!!

How to resolve?

by experiment ↓

$B \neq 0$

- There is no free current, no bounded current (no magnetic material), what is generating magn. field?
- From the exp.: B disappears, when the capacitor is ~~in static~~ fully charged, i.e. $\frac{dQ}{dt} = 0$ $(I = 0)$

Maxwell's observation: the magnetic field is

generated by the changing electrical field! ②

let's investigate the changing el. field in the capacitor.

$$|I| = \frac{dQ_c}{dt} = \frac{d\sigma}{dt} A \quad \text{in a capacitor}$$

$$|\underline{E}| = \frac{\sigma}{\varepsilon_0} \Rightarrow \sigma = \varepsilon_0 E$$

$$I = A \frac{d E \cdot \varepsilon_0}{dt} = \frac{d(\varepsilon_0 E A)}{dt} = \frac{d\Phi_D}{dt}$$

time derivative of the flux of dielectric displacement!

so for surface A_2: we have a conductive current I

for surface A_1: we have a changing electric field, $\frac{d\Phi_D}{dt} = I$ we call it displacement current!

* to corect Ampère's law, we have to add these two cuneuts:

$$\oint \underline{B} d\underline{r} = \mu_0 I + \mu_0 \frac{d\Phi_D}{dt} = \mu_0 I + \mu_0 \frac{d}{dt} \iint \varepsilon_0 \underline{E} d\underline{s}$$

generalized Ampère's law we will call it Amp-Maxw. law

if we have material:

$$\boxed{\oint \underline{H} d\underline{r} = I + \frac{d}{dt} \iint_{A_1 A_2} \underline{D} d\underline{s}} \quad \text{or} \quad \oint \underline{H} d\underline{r} = \iint \left(j + \frac{dD}{dt}\right) ds$$

Stokes' theorem:

$$\oint \underline{H}\,d\underline{l} = \iint_{A_1(A_2)} \operatorname{curl} \underline{H}\,d\underline{s} = \iint_{A_1} \underline{j} + \frac{d\underline{D}}{dt}\,d\underline{s}$$

$$\Updownarrow$$

$$\boxed{\operatorname{curl} \underline{H} = \underline{j} + \frac{d\underline{D}}{dt}}$$

Milestone!

(Marble memorial in Westminster abbey, next to Newton's tomb)

* Difficult question: We do not need a capacitor or a closed loop, B will be there wherever E is changing!!

Does the displacement current imply the physical motion of charge carriers ??

- Not in vacuum!
- Yes in dielectric material!

$$\underline{j}_{displ} = \frac{d\underline{D}}{dt} = \frac{d}{dt}(\varepsilon_0 \underline{E} + \underline{P}) =$$

$$= \underbrace{\varepsilon_0 \frac{d\underline{E}}{dt}}_{\text{vacuum}} + \underbrace{\frac{d\underline{P}}{dt}}_{\text{polarization current!}}$$

$$\left(\underline{P} = n\,\underline{d_e} = nq\underline{l} \qquad \frac{d\underline{P}}{dt} = nq\frac{d\underline{l}}{dt} = \underbrace{nq\underline{v_p}}_{\text{current density of polarizing bounded charges}}\right.$$

Maxwell's equations

(2)

Faraday's law $\oint_\Gamma \underline{E}\,d\underline{r} = -\frac{\partial}{\partial t}\left(\iint_A \underline{B}\,d\underline{s}\right)$

$\Updownarrow$

$$curl\,\underline{E} = (rot\,\underline{E}) = -\frac{\partial \underline{B}}{\partial t}$$

Meaning: ~~Electro~~static electric field is vortex-free (rotation free)

Induced el. field has vortices, its rotation strength is ~~propo~~ equal to the negative derivative of the flux of magnetic induction crossing the closed loop.

Differential "vortex density" of E is $-\frac{\partial B}{\partial t}$

(1) Ampère-Maxwell law

$$\iint \underline{H}\,d\underline{r} = I + \frac{d}{dt}\iint \underline{D}\,d\underline{s} \iff curl\,\underline{H} = \underline{j} + \frac{\partial \underline{D}}{\partial t}$$

Meaning:

Magnetic field is always composed of vortices, its rotation strength is equal to the conductive current + displacement current



Mutual coupling between $\underline{E}$ and $\underline{B}$

Gauss's law

$$\oint \underline{E}\,d\underline{s} = \frac{1}{\varepsilon_0} Q$$ in vacuum

$$\left(-\frac{1}{\varepsilon_0}\oint \underline{P}\,d\underline{s} \text{ for materials}\right)$$

$$\oint \underline{D}\,d\underline{s} = Q$$

$$\Updownarrow \text{ Gauss-Ostrogradsky}$$

$$\operatorname{div} \underline{D} = \varrho$$

Electric field has sources, the field lines of static electricity start and end on charges. Source strength of static electricity is proportional Flux of a closed surface to the enclosed charge.

Induced electricity does not appear here, since it is source-less

Gauss's law for magnetic field

$$\oint \underline{B}\,d\underline{s} = 0 \qquad \operatorname{div} \underline{B} = 0$$

Magnetic field is source-less, magnetic induction lines are always closed loops

$$\underline{D} := \varepsilon_0 \underline{E} + \underline{P}$$

$$B = \mu_0 (\underline{H} + \underline{M})$$

$$\underline{j} = \sigma \underline{E}$$

Derivation of the conservation of ~~energy~~ charge:

$$\text{div } \underline{D} = \varrho \quad \xrightarrow{\frac{d}{dt}} \quad \text{div } \frac{\partial \underline{D}}{\partial t} = \frac{\partial \varrho}{\partial t}$$

$$\text{curl } \underline{H} = \underline{j} + \frac{\partial \underline{D}}{\partial t}$$

$$\frac{\partial \underline{D}}{\partial t} = \text{curl } \underline{H} - \underline{j}$$

$$\text{div}(\text{curl } \underline{H}) - \text{div } \underline{j} = \frac{\partial \varrho}{\partial t}$$

$\text{div}(\text{curl } \underline{H})$ — $\varnothing$ (from math)

$$-\text{div } \underline{j} = \frac{\partial \varrho}{\partial t} \quad \text{(Gauss-O)}$$

$$-\iiint \text{div } \underline{j} \, dV = \iiint \frac{\partial \varrho}{\partial t} \, dV$$

$$\boxed{-\oint\!\!\oint \underline{j} \, d\underline{s} = \frac{\partial Q}{\partial t}}$$

